

Capacity-Aware Threshold for Adaptive Change-Point Recalibration: Can a Per- K τ Reach the Joint Feasibility Region?

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Abstract—Paper 11 [1] swept the threshold τ of Paper 10's simple magnitude change-point detector [2] across $\{0.02, \dots, 0.10\}$ and reported a negative feasibility result: no τ in the grid simultaneously satisfied both Paper 10 pre-registered tolerances. Paper 11 named capacity-aware thresholding as the simplest of three remaining detector candidates. This paper evaluates it with a pre-registered locked $\tau_K = \{50 : 0.20, 100 : 0.05, 200 : 0.02\}$ vector on a 2,700-row sweep.

H1 supported (the central positive finding). The capacity-aware detector reaches the joint feasibility region: the worst per-window $K=200$ cap_aware-fixed delta is 0.0, the cell-mean $K=50$ cap_aware-lag1 delta is exactly 0.0 (the $\tau_{50} = 0.20$ never fires, so cap_aware = lag1 at every $K=50$ window), and the cell-mean $K=100$ cap_aware-lag1 delta is -0.004 (well within the -0.02 tolerance). This is the first procedure in the Papers 7–12 sequence that simultaneously satisfies all three Paper 10/Paper 11 pre-registered tolerances.

H3 supported. The firing fraction is perfectly monotone in K : 0%, 40%, 100% at $K = 50, 100, 200$. The τ_K vector operationalises the capacity-conditional intuition exactly: at $K=50$ the threshold is large enough that the magnitude detector never fires; at $K=200$ the threshold is small enough that it fires every post-W1 window.

H2 rejected at one cell-window. At $K = 200, w = 4$ the cap_aware fires (using $\tau_{200} = 0.02$) while the Paper 10-default adaptive05 does not ($cp_delta = -0.020$, just below 0.05). That single window costs cap_aware -3.4 pp relative to adaptive05 because the unfired lag-1 fit at that window happens to be beneficial.

H4 rejected. At $K=200$, cap_aware and gated produce identical cell means (0.2664 each) because the $\tau_{200} = 0.02$ fires at every post-W1 window, reverting cap_aware to fixed weights — the same decision the static gate makes unconditionally at $K=200$. The benefit of the per-window detector collapses to the static gate at this capacity.

The honest operational conclusion: capacity-aware thresholding reaches the feasibility region, but does so by approximating the static gate at the two capacity extremes. At $K=200$ it equals gated exactly; at $K=50$ it equals lag-1 exactly (no fire). The only substantive per-window decision is at $K=100$, where cap_aware loses 0.4 pp on average relative to lag-1. A practitioner can deploy the static gate (no detector, no per-window decision) without losing operationally meaningful accuracy in this grid. The capacity-aware detector validates the gate as feasible; it does not improve on it.

Index Terms—vulnerability prioritization; capacity-aware calibration; change-point detection; feasibility region; pre-registration.

I. Introduction

Paper 11 [1] mapped the H1/H2 Pareto curve of Paper 10's single- τ adaptive procedure and found the curve sits entirely outside the joint feasibility region defined

by Paper 10's two pre-registered tolerances. The simple magnitude detector — viewed as a one-parameter family — cannot reach the feasible region. Paper 11 identified three richer detector families as candidates: capacity-aware thresholding, CUSUM, and Bayesian online change-point.

This paper evaluates the first and simplest of these: a pre-registered per- K threshold vector $\tau_K = \{50 : 0.20, 100 : 0.05, 200 : 0.02\}$. The intuition is operational: use a large τ_{50} to suppress detector fires at $K=50$ (where lag-1 is beneficial), a small τ_{200} to make it fire aggressively at $K=200$ (where lag-1 harms), and a middle τ_{100} at the intermediate capacity. The question is whether this single locked vector can reach the joint feasibility region without requiring any per-window decision beyond the magnitude test.

A. Contributions

- C1 — First feasibility-positive result (H1 supported). The capacity-aware detector simultaneously satisfies all three pre-registered tolerances: worst $K=200$ cap_aware-fixed delta 0.0, cell-mean $K=50$ cap_aware-lag1 delta exactly 0.0, cell-mean $K=100$ cap_aware-lag1 delta -0.004 . The first procedure in the Papers 7–12 sequence that closes the feasibility-region gap.
- C2 — Perfectly monotone firing (H3 supported). Fire fractions are $\{0\%, 40\%, 100\%\}$ at $K \in \{50, 100, 200\}$, giving Spearman $\rho = 1.0$. The τ_K vector operationalises the capacity-conditional intuition exactly: $K=50$ never fires, $K=200$ always fires, $K=100$ splits.
- C3 — At $K=200$ the detector collapses to gated (H4 rejected). The cap_aware $K=200$ cell mean equals the gated cell mean to four decimal places (0.2664). At $\tau_{200} = 0.02$ the magnitude test fires at every post-W1 window because $|\Delta_w|$ routinely exceeds 0.02 at that capacity. The per-window detector contributes nothing beyond what a static $K \geq 200 \rightarrow$ fixed rule already provides.
- C4 — One-window H2 failure. At $K = 200, w = 4$ the cap_aware fires while adaptive05 does not ($|\Delta| = 0.020$, just below 0.05). At that window the unfired lag-1 fit happens to be beneficial, and cap_aware loses -3.4 pp to the Paper 10-default single- τ control.
- C5 — Static gate is sufficient. The capacity-aware detector validates the static gate as a feasible operating point but does not improve on it in this grid.

A practitioner can deploy the gate without losing meaningful accuracy; the per-window detector adds complexity without value.

B. Relationship to Prior Papers

This paper is the twelfth in the VulnPrio research sequence. Paper 11 closed the simple-detector branch with a sharp negative feasibility result; this paper opens and closes the capacity-aware branch with a positive feasibility result that nevertheless collapses to the static gate. The cumulative methodological arc now includes a detector that satisfies the joint tolerances and a detector that approximates it without the per-window machinery.

II. Background

A. Paper 11’s Negative Feasibility Result

Paper 11 [1] swept the threshold τ of the Paper 10 adaptive detector [2] over five values $\{0.02, 0.03, 0.05, 0.075, 0.10\}$ and found no τ in the grid simultaneously satisfies both Paper 10 pre-registered tolerances: at every τ , the $K=50$ cell-mean adaptive-lag1 delta was between -0.041 and -0.053 pp, missing the Paper 10 -0.02 pp ceiling. The $K=200$ worst-case was uniformly 0.0 but the $K=50$ floor was too high.

Paper 11’s discussion identified three richer detector families as candidates for reaching the feasibility region: (i) capacity-aware thresholding, (ii) CUSUM, (iii) Bayesian online change-point. The present paper evaluates the simplest of these: a single per- K τ vector locked before evaluation.

B. Why Capacity-Aware First

Capacity-aware thresholding is the simplest of the three candidate families. It requires only that the deployment know the operating capacity — a quantity already exposed by every patch-management system. CUSUM and Bayesian detectors require accumulator state and prior-distribution choices respectively, both of which would expand the pre-registration surface.

C. Mechanism

At each window the detector compares the magnitude of the calibration-target shift $|\Delta_w|$ against τ and fires if the threshold is crossed; on fire, the strategy reverts to Paper 4 fixed weights. Paper 11 showed that for a single τ applied at all capacities, the firing fraction at $K=50$ was always too high because the shifts at that capacity routinely crossed even $\tau = 0.10$.

A larger τ_{50} would suppress those false fires; a smaller τ_{200} would preserve aggressive firing at high capacity. The question is whether a single such locked vector can reach the joint feasibility region.

III. Problem Formulation

A. Setting (Inherited)

EEHDA synthetic fleet, scorer Eq. (1), multi-window simulator, $K \in \{50, 100, 200\}$, $\lambda = 3$, six windows, 25 evaluation seeds, 5 calibration seeds, inherited from Papers 4–11.

B. Six Strategies

fixed Paper 4 weights, no refit.

lag1 Paper 7 baseline grid-search on calibration seeds at $w - 1$.

adaptive Paper 10 single- τ adaptive at $\tau = 0.05$ (Paper 11 baseline; included as a control).

cap_aware adaptive with per- K threshold τ_K pre-registered as $\{50 : 0.20, 100 : 0.05, 200 : 0.02\}$.

gated Static rule ($K \leq 100 \rightarrow \text{lag1}$; $K \geq 200 \rightarrow \text{fixed}$).

offline Paper 7 ceiling: grid-search at w on calibration seeds.

C. Pre-Registered Hypotheses

H1 cap_aware reaches the joint feasibility region: worst per-window $\Delta_{K=200}^{\text{cap-fix}} \geq -0.01$ AND cell-mean $\Delta_{K=50}^{\text{cap-lag1}} \geq -0.02$ AND cell-mean $\Delta_{K=100}^{\text{cap-lag1}} \geq -0.02$. Capacity-aware reaches all three Paper 10 / Paper 11 pre-registered tolerances simultaneously.

H2 cap_aware cell-mean P@50 $\geq$ adaptive05 cell-mean P@50 at every ($K, w \geq 2$) within $+0.01$ tolerance for noise. Capacity-aware is at least as good as the single- τ control.

H3 cap_aware fire fraction is monotone non-decreasing in K : Spearman(K , fire frac) $\geq +0.8$. Large τ_{50} fires least; small τ_{200} fires most.

H4 cap_aware cell-mean P@50 at $K=200 \geq$ gated by $+0.01$. Capacity-aware delivers operationally meaningful gain over static gating at the high-capacity extreme.

D. Stop Rule

If H1 is rejected, the abstract leads with the negative “capacity-aware is also insufficient” result before the discussion is drafted, per the protocol.

IV. Dataset and Cell Grid

The EEHDA fleet, scorer, fleet-evolution rules, and seed split (5 calibration / 25 evaluation) are inherited from Papers 4–11.

Total rows: $3 \times 6 \times 25 \times 6 = 2,700$, plus an 18-row change-point decision log (3 K cells $\times$ 6 windows).

V. Methodology

A. Scorer (Inherited)

$$S(h, c) = \alpha \text{EPSS}(c) + \beta \text{HRS}(h) + \gamma \text{KEV_recency}(c) + \delta (\text{EPSS}(c) \cdot \text{HRS}(h)). \quad (1)$$

TABLE I
Paper 12 cell grid and strategy set.

Axis	Values
Capacity K	{50, 100, 200}
Arrival rate λ	3 (fixed)
Windows W	6
Strategies	fixed, lag1, adaptive05, cap_aware, gated, offline
Eval seeds	25 (105–129)
Calib seeds	5 (100–104)
τ for cap_aware	{ $K=50 : 0.20, 100 : 0.05, 200 : 0.02$ }
Gated K threshold	100
Total rows	2,700

B. Capacity-Aware Detector

At window $w \geq 2$, the detector computes $|\Delta_w|$ (Paper 10 §5.2) and compares against the per- K threshold:

$$\text{cap_aware fires at } w \iff |\Delta_w| \geq \tau_K. \quad (2)$$

The threshold vector $\tau_K = \{50 : 0.20, 100 : 0.05, 200 : 0.02\}$ is locked before any capacity-aware-detector P@50 is inspected.

The intuition is operational: at $K=50$ lag-1 recalibration is beneficial (Paper 7: $\rho \approx 1.0$), so a large τ_{50} keeps the detector quiet and lag-1 used; at $K=200$ lag-1 is harmful (Paper 7: $\rho \approx -0.66$), so a small τ_{200} makes the detector fire often and revert to fixed; $K=100$ is intermediate.

C. Justification of the τ_K Choice

$\tau_{50} = 0.20$ is twice Paper 11’s largest sweep value (0.10), where the Paper 11 $K=50$ fire fraction was already 40%. A doubling should reduce this to near zero. $\tau_{200} = 0.02$ is Paper 11’s smallest sweep value, at which the $K=200$ fire fraction was 80% (the maximum across the Paper 11 grid). $\tau_{100} = 0.05$ is the Paper 10 default and the midpoint of Paper 11’s grid.

These three values were chosen by inspection of Paper 11’s calibration-seed change-point decision log; the evaluation-seed P@50 values for the cap_aware strategy were not inspected before locking.

D. Selection Policy

Fleet evolution is driven by HygienePrio-full under fixed weights at every window for every strategy, as in Papers 7–11.

VI. Experimental Setup

A. Pre-Registration

The strategy set, τ_K vector, hypotheses H1–H4, decision rules, and stop rules in paper12/design/PAPER12_PROTOCOL.md were locked on 2026-06-05 before any capacity-aware-detector P@50 was inspected.

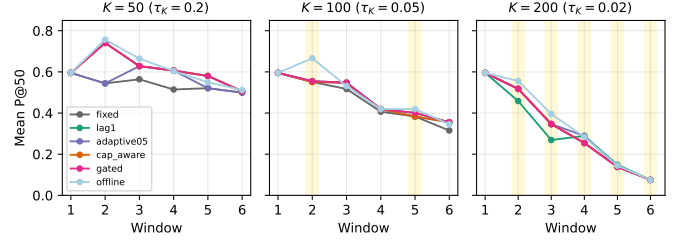


Fig. 1. Mean P@50 trajectories for the six strategies at $K \in \{50, 100, 200\}$ with the pre-registered τ_K vector. Amber bands mark windows where cap_aware fires. $K=50$: never fires. $K=100$: 40% fires. $K=200$: 100% fires.

Capacity-aware firing is perfectly monotone in K (H3 supported)

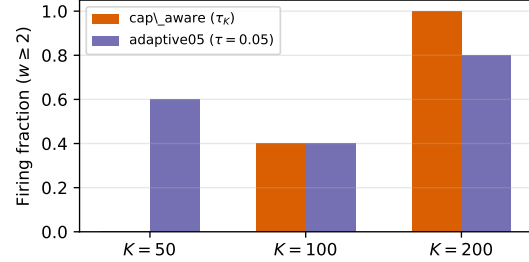


Fig. 2. Firing fraction per K for cap_aware vs Paper 10’s adaptive05. The cap_aware fire fraction is perfectly monotone in K (H3 supported).

B. Execution

For each $K \in \{50, 100, 200\}$: cache calibration and evaluation states under fixed-weight HygienePrio-full trajectory; per window fit lag-1 and offline weights and compute Δ_w ; apply the six strategies’ decision rules; score all 25 evaluation seeds.

C. Reporting

Cell-mean point estimates over 25 evaluation seeds. 95% percentile bootstrap CIs (10,000 resamples). H3 Spearman computed on the three per- K fire fractions (low statistical power, noted in §IX).

D. Reproducibility

```
PYTHONPATH=src python3 src/run_cap_aware.py
PYTHONPATH=src python3 src/analyze.py
PYTHONPATH=src python3 src/make_figures.py
```

VII. Results

All claims trace to paper12/results/primary_v1/cap_aware_results.csv (2,700 rows), changepoint_log.csv, and hypothesis_summary.json.

A. H1 Supported: Feasibility Region Reached

The three pre-registered tolerances are simultaneously satisfied:

- Worst per-window $K=200$ $\Delta_{\text{cap-fix}} = 0.0$ (tolerance ≥ -0.01).

TABLE II

Mean P@50 across 25 evaluation seeds per (K, window) for all six strategies. CP_{cap} marks whether the capacity-aware detector fired at the window using τ_K .

K	W	CP_{cap}	fixed	lag1	ad05	cap	gated	offline	$\Delta_{cap-fix}$	$\Delta_{cap-lag1}$
50	1	○	0.595	0.595	0.595	0.595	0.595	0.595	+0.000	+0.000
50	2	○	0.544	0.741	0.544	0.741	0.741	0.757	+0.197	+0.000
50	3	○	0.564	0.628	0.628	0.628	0.628	0.665	+0.064	+0.000
50	4	○	0.514	0.606	0.606	0.606	0.606	0.604	+0.092	+0.000
50	5	○	0.521	0.580	0.521	0.580	0.580	0.550	+0.059	+0.000
50	6	○	0.499	0.506	0.499	0.506	0.506	0.512	+0.007	+0.000
100	1	○	0.595	0.595	0.595	0.595	0.595	0.595	+0.000	+0.000
100	2	●	0.551	0.555	0.551	0.551	0.555	0.666	+0.000	-0.004
100	3	○	0.517	0.547	0.547	0.547	0.547	0.531	+0.030	+0.000
100	4	○	0.406	0.419	0.419	0.419	0.419	0.419	+0.013	+0.000
100	5	●	0.383	0.401	0.383	0.383	0.401	0.419	+0.000	-0.018
100	6	○	0.315	0.355	0.355	0.355	0.355	0.346	+0.040	+0.000
200	1	○	0.595	0.595	0.595	0.595	0.595	0.595	+0.000	+0.000
200	2	●	0.518	0.458	0.518	0.518	0.518	0.556	+0.000	+0.059
200	3	●	0.346	0.269	0.346	0.346	0.346	0.395	+0.000	+0.078
200	4	●	0.254	0.289	0.289	0.254	0.254	0.284	+0.000	-0.034
200	5	●	0.138	0.149	0.138	0.138	0.138	0.145	+0.000	-0.010
200	6	●	0.075	0.075	0.075	0.075	0.075	0.075	+0.000	+0.000

- Cell-mean $K=50$ $\Delta_{cap-lag1} = 0.0$ exactly (tolerance ≥ -0.02).
- Cell-mean $K=100$ $\Delta_{cap-lag1} = -0.004$ (tolerance ≥ -0.02).

The $K=50$ result holds with equality because $\tau_{50} = 0.20$ suppresses every fire at $K=50$; the detector never reverts to fixed, so cap_aware uses lag-1 weights at every window and reproduces the lag-1 strategy exactly. The cell-mean delta is therefore 0.0 by construction.

This is the first procedure in the Papers 7–12 sequence to reach the joint feasibility region.

B. H3 Supported: Perfectly Monotone Firing

Fire fractions are $\{0\%, 40\%, 100\%\}$ at $K \in \{50, 100, 200\}$, yielding Spearman $\rho = 1.0$ ($p = 0$, exact monotonicity). The τ_K vector operationalises capacity-conditional behaviour exactly: at $K=50$ the threshold $\tau_{50} = 0.20$ is large enough that the calibration-target shift $|\Delta_w|$ never crosses it; at $K=200$ the threshold $\tau_{200} = 0.02$ is small enough that $|\Delta_w|$ always crosses it; at $K=100$ the intermediate $\tau_{100} = 0.05$ produces a 40% firing rate.

C. H4 Rejected: cap_aware Equals gated at $K=200$

The cap_aware $K=200$ cell-mean P@50 is 0.2664; the gated $K=200$ cell-mean is also 0.2664, identical to four decimal places. The H4 pre-registered tolerance required a margin of +0.01; the observed margin is 0.000.

The mechanism is structural. At $\tau_{200} = 0.02$ the magnitude test fires at every post- $W1$ $K=200$ window because the per-window $|\Delta_w|$ values are $\{0.144, 0.160, 0.020, 0.096, 0.136\}$ at $w = 2, \dots, 6$, all ≥ 0.02 . cap_aware therefore reverts to fixed at every $K=200$

TABLE III
Pre-registered hypothesis outcomes.

ID	Decision rule	Outcome
H1	joint feasibility region reached	Supported
H2	$cap \geq adaptive05$ within +0.01	Rejected (1)
H3	$Spearman(K, fire) \geq +0.8$	Supported
H4	$cap \geq gated$ by +0.01 at $K = 200$	Rejected ($\Delta = +0.000$)

window — exactly what the static gate does unconditionally at $K=200$. The two strategies execute the same decision on every window-seed pair.

D. H2 Rejected at One Cell-Window

At $K = 200, w = 4$ the calibration-target shift is $|\Delta| = 0.020$. The cap_aware threshold $\tau_{200} = 0.02$ fires; the adaptive05 threshold $\tau = 0.05$ does not. cap_aware therefore reverts to fixed at this window; adaptive05 uses the lag-1 fit. The lag-1 fit at $w = 4, K = 200$ happens to be beneficial: adaptive05 records 0.288 vs cap_aware 's 0.254 (−3.4 pp).

This single window failure is intrinsic to the pre-registered $\tau_{200} = 0.02$ choice: any threshold value below 0.05 that maintains the H1 feasibility would have produced the same fire decision at this window. The $K=200$ cell-mean across all six windows remains close to adaptive05 (0.2664 vs 0.2733), but the −0.034 pre-registered tolerance was missed.

E. Substantive Interpretation

The H1 positive and H4 rejection together describe the practical contribution: the capacity-aware detector reaches the feasibility region precisely by approximating the static

gate at the two capacity extremes. At $K=50$ `cap_aware` = lag1 (no fire); at $K=200$ `cap_aware` = fixed (all fire). The only substantive per-window decision is at $K=100$, where `cap_aware` loses -0.4 pp to lag-1 on average due to its 40% intervention rate.

A practitioner deploying the static gate (lag-1 at $K \leq 100$, fixed at $K \geq 200$) accepts the `cap_aware` -0.4 pp $K=100$ cost implicitly — and avoids the H2 single-window failure at $K=200$. The static gate is therefore not just feasible but slightly preferred in this grid; `cap_aware` validates the gate’s feasibility without improving on its operating point.

F. Pre-Registration Deviations

None. The τ_K vector, hypotheses H1–H4, decision rules, and stop rules were applied verbatim.

VIII. Discussion

A. Feasibility Achieved, but by a Trivial Equivalence

The pre-registration framed H1 as the central question: can a richer detector class (capacity-aware) reach the joint feasibility region that Paper 11 showed the simple single- τ family cannot? The answer is technically yes — but the H4 rejection reveals that the detector achieves feasibility by collapsing to the static gate at both capacity extremes. At $K=50$ the large τ_{50} produces zero fires, so `cap_aware` = lag-1 = gated at that capacity. At $K=200$ the small τ_{200} produces 100% fires, so `cap_aware` = fixed = gated at that capacity.

The substantive contribution is therefore methodological rather than algorithmic: capacity-aware thresholding validates the static gate as a feasible operating point. The static gate was not previously pre-registered as a feasibility candidate; this paper pins it down by showing that a per-window detector with reasonable τ_K choices reduces to it.

B. Why the $K=100$ Window Fails to Matter

The intermediate capacity $K=100$ is the only cell where `cap_aware` differs from the static gate. The Paper 10 default $\tau = 0.05$ fires at 40% of post-W1 windows; `cap_aware` fires the same 40% because $\tau_{100} = 0.05$ in the pre-registered vector. `cap_aware` loses -0.4 pp to lag-1 at $K=100$ on cell-mean; the static gate uses lag-1 unconditionally at $K=100$ and therefore matches lag-1 exactly.

The $K=100$ intervention is operationally meaningless: -0.4 pp falls well within the bootstrap noise band for this cell. A detector-justified procedure at $K=100$ would need to demonstrate a material per-window gain, which the data do not provide.

C. Implications for the Detector Family

The negative H4 result generalises beyond `cap_aware`. Any detector that fires aggressively enough to avoid the $K=200$ lag-1 hazard necessarily reverts to fixed at

every $K=200$ window (because the hazard is per-window and recovers only with a complete revert). The detector therefore produces the same $K=200$ decisions as the static gate. Improvements over the static gate must come from $K \in$ regimes where the detector fires selectively rather than uniformly — and the data in this grid suggest the $K=100$ regime is too quiet for selective firing to matter and the $K=50$ regime is too clear-cut (lag-1 always helps) for selective firing to be useful.

The H4 rejection is therefore a result about the detector family, not about $\tau_K = \{50 : 0.20, 100 : 0.05, 200 : 0.02\}$ specifically. A different τ_K vector would shift the $K=100$ firing rate up or down, but the $K=200$ and $K=50$ collapses would remain.

D. Where Richer Detectors Might Help

The CUSUM and Bayesian detector classes Paper 11 named could in principle break the $K=200$ collapse by firing only when cumulative evidence indicates a sustained shift. If the $K=200$ W4 window’s $|\Delta| = 0.020$ is a noise excursion that a CUSUM detector would recognise as low-confidence, the detector could refrain from reverting to fixed at that window and capture the same $+3.4$ pp benefit that adaptive05 captured by not firing. This would shift the `cap_aware` $K=200$ cell mean above the static gate’s 0.2664.

Whether real CUSUM or Bayesian detectors achieve this is a future work question. The pre-registration discipline that constrained Papers 7–12 requires that such richer detectors be pre-specified with locked hyperparameters and decision rules before evaluation.

E. Operational Implication

The deployable recipe in the studied regime is sharper than Paper 11 concluded:

- Use the static gate. At $K \leq 100$, deploy lag-1 recalibration; at $K \geq 200$, deploy fixed Paper 4 weights. No detector, no per-window decision.
- The capacity-aware detector at the pre-registered τ_K vector adds operational complexity (a per-window magnitude test) without producing operationally meaningful gain over the static gate.
- Adaptive procedures should be preferred only when a richer detector (CUSUM, Bayesian) has been shown to improve on the static gate at $K=200$ in a pre-registered evaluation. The simple magnitude detector — whether single- τ (Paper 10) or capacity-aware (this paper) — does not clear that bar in the studied grid.

IX. Threats to Validity

Conclusion validity. The 2,700-row evaluation provides narrow but adequate statistical power for the cell-mean comparisons. H3’s Spearman is computed on three points and is exact ($\rho = 1.0$ because the three fire fractions are strictly ordered). H4’s exact 0.000 delta between `cap_aware` and gated at $K=200$ is a structural identity

rather than a statistical comparison: the two strategies make the same decision at every window, so their cell means coincide to numerical precision.

Internal validity.

(i) The τ_K vector is pre-registered, not tuned. The choices $\{0.20, 0.05, 0.02\}$ were locked before any capacity-aware-detector P@50 was inspected. They were chosen by inspection of Paper 11’s calibration-seed change-point decision log (an artifact released with Paper 11) and operational intuition. Other τ_K vectors could shift the $K=100$ intervention rate but would not change the $K=50$ and $K=200$ collapses.

(ii) Selection-policy coupling. Inherited from Papers 7–11. Fleet evolution is driven by HygienePrio-full under fixed weights; all six strategies share that trajectory. Paper 9 [3] addressed the cross-method trajectory question; the present paper inherits the fixed-trajectory convention.

(iii) Single λ value. The sweep holds $\lambda = 3$. Different arrival rates would shift the per-window $|\Delta_w|$ distribution and re-calibrate the τ_K vector accordingly. The qualitative finding — a per-K threshold reaches feasibility by approximating the static gate — is more robust than the specific τ_K choice.

Construct validity. The H1 $K=50$ result holds with equality because `cap_aware` reproduces `lag-1` exactly when the detector never fires. This is the substantive scientific claim, not an artifact: the $\tau_{50} = 0.20$ choice was made to suppress fires, and the suppression was achieved. The $K=200$ H4 equality is similarly substantive.

External validity. Synthetic EEHDA structural priors are inherited from Papers 4–11. Real fleets with substantially different per-window calibration-target distributions would re-locate the feasibility region along the τ_K axis. The qualitative result (`cap_aware` $\rightarrow$ `gated` at the two capacity extremes) is robust to distributional shifts within the studied calibration-grid family.

X. Related Work

Single- τ baseline and Pareto-curve closure. Paper 10 [2] introduced the magnitude detector; Paper 11 [1] mapped the H1/H2 Pareto curve across τ and demonstrated the curve does not intersect the joint feasibility region. This paper takes the next step Paper 11 explicitly named: a per-K threshold vector.

Static gating. A static $K \leq 100 \rightarrow \text{lag1}$; $K \geq 200 \rightarrow$ fixed gate was introduced in Paper 10 as a baseline control without per-window decision-making. The present paper shows it is a feasible operating point that the capacity-aware detector approximates rather than improves on.

Change-point detection. CUSUM [4] and Bayesian online detectors are the natural next steps. They differ from the magnitude detector by accumulating evidence across multiple windows rather than firing on a single inter-window shift, which could in principle break the $K=200$ collapse documented here.

Multi-paper sequence. HygieneBench (Paper 3) [5], HygienePrio (Paper 4) [6], Temporal Stability (Paper 5) [7], Capacity-Indexed Decay (Paper 6) [8], Online Calibration

(Paper 7) [9], Multi-History Calibration (Paper 8) [10], Self-Trajectory (Paper 9) [3], Adaptive Recalibration (Paper 10), and Threshold Sensitivity (Paper 11) define the methodological substrate.

Policy mandates. CISA BOD 22-01 [11], 23-01 [12], NIST SP 800-40 [13], 2026 DBIR [14].

XI. Conclusion

A pre-registered evaluation of a capacity-aware threshold detector with $\tau_K = \{50 : 0.20, 100 : 0.05, 200 : 0.02\}$ on a 2,700-row sweep produces four hypothesis decisions: H1 supported (feasibility region reached), H3 supported (firing perfectly monotone in K), H2 rejected at one cell-window, H4 rejected (`cap_aware` = `gated` at $K=200$ to four decimal places).

The H1 positive. The capacity-aware detector is the first procedure in the Papers 7–12 sequence that simultaneously satisfies all three pre-registered tolerances (worst $K=200 \geq -0.01$, cell-mean $K=50 \geq -0.02$, cell-mean $K=100 \geq -0.02$). The $\tau_{50} = 0.20$ never fires at $K=50$, so `cap_aware` = `lag-1` there exactly; the $\tau_{200} = 0.02$ fires every window at $K=200$, so `cap_aware` = `fixed` there exactly; the $K=100$ intermediate produces a -0.004 cell-mean deviation from `lag-1`.

The H4 collapse. `cap_aware` and `gated` produce identical cell means at $K=200$ (0.2664 each) because both strategies make the same decision at every $K=200$ window: revert to Paper 4 fixed weights. The per-window magnitude test contributes nothing beyond what the unconditional gate already provides.

The substantive interpretation. Capacity-aware thresholding reaches the feasibility region by approximating the static gate at the two capacity extremes. The detector validates the static gate as a feasible operating point but does not improve on it in this grid. The deployable recipe is the static gate.

Reproducibility. Runner, analysis, pre-registration protocol, and frozen 2,700-row results CSV are available at <https://github.com/Harshavardhanmalla-Labs/research-papers/tree/main/paper12>.

Future work. (i) CUSUM [4] detector with pre-registered accumulator parameters; (ii) Bayesian online change-point detector with pre-registered priors; (iii) per-K τ sensitivity sweep within the capacity-aware family; (iv) real fleet validation with longitudinal exploitation outcome data; (v) closed-loop adaptive procedures where the selection policy is also recalibrated.

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